they are presented with. *Task-agnostic learning* expands on the scenario of Domain-Incremental Learning by eliminating the use of task IDs during the training phase (Zeno et al. [14]). A task-agnostic agent is capable of learning in an environment without any extrinsic goal [8], meaning no specific reward function is given during training. Specifically, this means that the agent can solve different tasks without knowing their boundaries or task-IDs, i.e. indicating which task they are solving.

4 Task-Agnostic Policy Distillation

In knowledge distillation [17, 30], the goal is to train a target network, referred to here as the knowledge base, to produce the same output distribution as the original network, referred to here as the active column. In the proposed task-agnostic phase, the active column is initially trained to maximize an intrinsic reward that encourages exploration, without relying on task-specific extrinsic rewards. Subsequently, the knowledge from the active column is transferred to the knowledge base by distilling the exploratory behavior of the trained active column network into the knowledge base network. This process is repeated iteratively as necessary. Below, we provide more details on the training environment and the variations of the task-agnostic phase, followed by a discussion on the intrinsic reward used for training.

4.1 Meta-Environment

Let $\mathcal{E}$ represent the set of all environments. For a given task k, let $E^k \in \mathcal{E}$ denote the environment E with task k, where $k \in \mathbb{N}$. Within the context of continual learning, specifically in the Atari 2600 domain [16], it is often assumed that each environment is considered a distinct task. This concept extends to the idea of learning task k within a given environment E and $r_{t+1}, s_{t+1} \sim P^{E^k \in \mathcal{E}}(\cdot)$ represents the dynamics of that environment. This formalism is congruent and applicable to scenarios where different tasks exist within a single environment, such as a robot solving different tasks in an environment, where E^k signifies a task within E. In the case of the Atari 2600 domain [16], we consider all games/environments to constitute one unified environment and conceptualize $\mathcal{E}$ as a Meta-Environment, dependent upon the task context and the definition of what constitutes a task, given the versatile nature of the term task. The Meta-Environment replicates a condition where a single environment is present, encompassing diverse tasks. This concept is consistently applied throughout the paper, presenting inherent robustness to changes in environmental dynamics, such as the sudden introduction of a new game, which are interpreted as shifts within the Meta-Environment and therefore do not negatively impact the task-agnostic policy (see Section 4.2). For a visual representation, see Fig. 1. It is important to note that the tasks can be sampled in a predefined, specified sequence from the Meta-Environment. However, this specified sequence can be relaxed to represent a specific distribution, from which the tasks (game environments) are sampled. Learning within the Meta-Environment can be more challenging due to the different textures of the tasks (i.e. game environments) and the high dissimilarity between tasks.

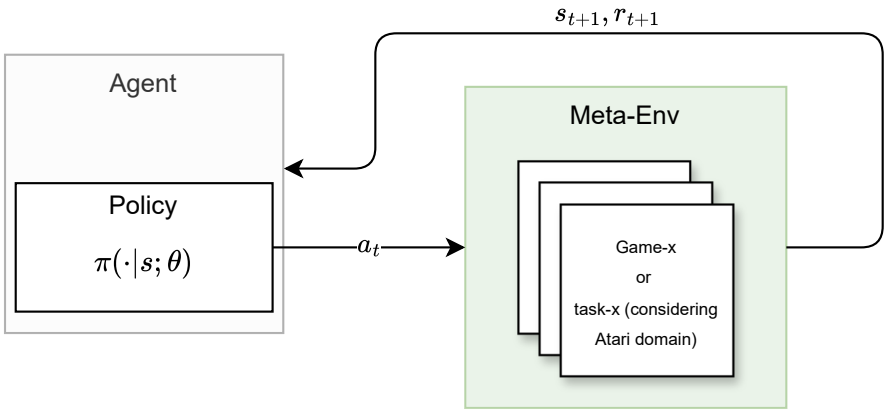

Fig. 1: Illustration of the Meta-Environment, representing different game environments as tasks within the context of Atari 2600 games.

4.2 Variations of the Task-Agnostic Phase

As proposed by Schwarz et al. [13], the generality of the introduced progress and compress algorithm makes it a suitable candidate for integration into various frameworks. The work of Schwarz et al. [13] is extended to include a *task-agnostic* phase. Consequently, this paper introduces the *Task-Agnostic Policy Distillation* framework, which augments the progress and compress framework. Building on the foundational architecture of Schwarz et al. [13], the task-agnostic phase introduces variations that enable the agent to operate without explicit task boundaries [14]. This phase implies a learning period where the agent learns without an external goal, meaning it receives no rewards from the environment, addressing the question of learning without a well-structured reward function. A curiosity-driven intrinsic reward signal is introduced through self-supervised prediction (see Section 4.3), as conducted by [7, 15]. The task-agnostic phase develops its exploratory policy by maximizing the RL objective $\mathbb{E}_\pi \left[\sum_{n=0}^{\infty} \gamma^n r^i_{t+n}\right]$, where r^i_t represents an intrinsic reward generated by the agent itself at timestep t. This increases the probability of discovering novel states and potentially acquiring novel skills. The agent can leverage its exploratory policy and the generalized skills it induces to become a versatile problem solver when faced with specific tasks, addressing issues of positive forward transfer and enhancing sample efficiency. Specifically, the loss to be minimized in the task-agnostic phase is therefore:

$$\mathcal{L}^{agnostic}(\theta) = -\mathbb{E}_\pi \left[\sum_{n=0}^{\infty} \gamma^n r^i_{t+n}\right]. \tag{3}$$

Variation 1 (TAPD) Variation 1 of the task-agnostic phase is the most versatile among all the variations discussed in this work. It is suitable for diverse environments

in which the action space remains consistent but the tasks vary, where the environments are conceptualized as tasks by the Meta-Environment, a collection of different game environments. In this variation, tasks (game environments) are sampled based on a specific distribution from the Meta-Environment, such as uniform sampling here. This selection mechanism implies that the agent lacks awareness of when task transitions occur, reflecting the task-agnostic scenario outlined by Zeno et al. [14]. As depicted in Fig. 2, this variation builds upon the framework established by Schwarz et al. [13]. The architectural designs within the progress and compress phases align with the baseline [13]. The task-agnostic phase also makes use of the active column. The first variation of the task-agnostic phase can be described as a process that leverages the A2C algorithm to minimize $\mathcal{L}^{agnostic}(\theta)$. In this phase, the exploratory policy, i.e. the task-agnostic policy, is distilled into the knowledge base (KB) after a specified number of intermediate timesteps, followed by the selection of another game environment, highlighting the agent's unawareness of task transitions. Thus, this variation of the task-agnostic phase encompasses learning through both curiosity and distillation (see part (a) in Fig. 2). The distilled task-agnostic policy is then used through the lateral connections to learn a new task-agnostic policy. This new policy is once again distilled and used to enhance the knowledge base with elements of curiosity. The act of distilling the exploratory distribution of one policy into another augments the exploratory behavior, as distillation has more information per training sample than hard targets [30]. The integration of various task-agnostic policies into the knowledge base is performed in an alternating manner, i.e.

$$\xrightarrow[x-steps]{\text{explore/train}} \min \mathcal{L}^{agnostic}(\theta) \xrightarrow[x-steps]{\text{compress}} \theta^{kb} \xrightarrow[x-steps]{\text{explore/train}} \min \mathcal{L}^{agnostic}(\theta) \xrightarrow[x-steps]{\text{compress}} ... \tag{4}$$

where θ and θ^{kb} are the learning parameters of the active column and knowledge base, respectively.

This process is illustrated by the loop in Fig. 2, within the abstraction of the task-agnostic phase (see part (a) in Fig. 2). It is noteworthy that the Online EWC algorithm is employed throughout the process, ensuring the retention of knowledge from previous exploratory policies. Since the task-agnostic phase solely relies on the intrinsic reward r_t^i for learning, the agent does not require specific reward schemas, eliminating the need for any task-ID during the training process. This phase is suitable for sparse-reward environments as it does not require extrinsic rewards, making it an ideal pre-training step for downstream tasks. Once exploration is completed for a given number of steps, the Progress and Compress algorithm proposed by Schwarz et al. [13] is used for tasks that are designed to maximize extrinsic rewards (see part (b) in Fig. 2). The progress phase uses generalized knowledge from task-independent strategies to accomplish specific tasks. An analogy for this task-agnostic phase can be made with a child exploring different rooms. After exploring a room, the child consolidates the new experiences and knowledge gained (distillation). By exploring various rooms and consolidating their understanding each time they enter a new one, the child is subsequently assigned specific tasks by their parents.